

## Chapter 4 The statistical interpretation of entropy

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# Summary of Chapter 3

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1. The processes taken by a system undergoing a change of state can be classified into two types: *reversible* and *irreversible*.
  - When the change in the state of the system occurs as the result of the application of **a finite driving force**, the process proceeds irreversibly, and the degree of irreversibility of the process increases with the increasing magnitude of the driving force.
  - For a process to occur reversibly, the driving force must be **infinitesimal**, and thus, a reversible process proceeds at an infinitesimal rate. The system is imagined to move through a continuum of equilibrium states during a reversible process.

# Summary of Chapter 3

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**2.** When a system undergoes a change of state, during which it performs work and absorbs thermal energy, the magnitudes of the quantities  $w$  and  $q$  are maxima ( $w_{\max}$  and  $q_{\text{rev}}$ ), respectively, when the change of state occurs reversibly. For an irreversible path between the two states, less work is performed by the system, and correspondingly less thermal energy is absorbed.

**3.** There exists a state function called entropy,  $S$ , which can be written as

$$dS = \frac{\delta q_{\text{rev}}}{T}$$

when  $\delta q_{\text{rev}}$  enters (or leaves) the system. The difference between the entropy in state  $B$  and that in state  $A$  is thus

$$\Delta S = S_B - S_A = \int_A^B \frac{\delta q_{\text{rev}}}{T}$$

## Summary of Chapter 3

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4. If, in moving between the two states, the temperature of the system remains constant, the change in the entropy of the system is  $\Delta S = q_{\text{rev}}/T$  where  $q_{\text{rev}}$  is the thermal energy absorbed or given off by the system in moving *reversibly* between the two states.

5. If  $q_{\text{rev}}$  is provided by a constant-temperature heat reservoir at the temperature  $T$ , the entropy of the reservoir decreases by the amount  $q_{\text{rev}}/T$  as a result of the system moving from  $A$  to  $B$ . The entropy of the combined system plus heat reservoir is thus unchanged as a result of the *reversible process*; entropy has simply been transferred from the heat reservoir to the system.

## Summary of Chapter 3

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6. If the changes in the state of a system from  $A$  to  $B$  were carried out *irreversibly*, then less thermal energy,  $q$  ( $q < q_{\text{rev}}$ ), would be withdrawn from the heat reservoir by the system. Thus, the magnitude of the decrease in the entropy of the reservoir would be smaller (equal to  $q/T$ ). However, since entropy is a state function, the change in entropy of the gas,  $S_B - S_A$ , is independent of the process path, and  $\Delta S_{\text{system}} + \Delta S_{\text{heat reservoir}} > 0$ . Entropy has been created as a result of the occurrence of an irreversible process. The entropy created is termed  $\Delta S_{\text{irr}}$ .

## Summary of Chapter 3

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7. In the general case,  $S_B - S_A = q/T + \Delta S_{\text{irr}}$ , and as the degree of irreversibility increases, the thermal energy,  $q$ , withdrawn from the heat reservoir decreases and the magnitude of  $\Delta S_{\text{irr}}$  increases.
8. The increase in entropy, due to the occurrence of an irreversible process, arises from the degradation of the energy of the system, wherein some of the internal energy, which is potentially available for doing useful work, is degraded.

# Summary of Chapter 3

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**9.** A process occurring in an adiabatically contained system of constant volume (i.e., a system of constant  $U$  and  $V$ ) will proceed irreversibly with a consequent production of entropy until the entropy is *maximized*. The attainment of maximum entropy is the criterion for equilibrium. Thus, the entropy of an adiabatically contained system can never decrease; it increases as the result of an irreversible process and remains constant at its maximum value during a reversible process.

**10.** A combination of the First and Second Laws of Thermodynamics gives, for a closed system which does no work other than the work of expansion against a pressure,  $dU = TdS - PdV$ . For  $U$ , the natural choice of the independent variables is  $S$  and  $V$ .

## 3.13 Maximum work

The Second Law:  $dS_{\text{system}} = \frac{\delta q}{T} + dS_{\text{irr}}$

The First Law:  $\delta q = dU_{\text{system}} + \delta w$

$$\left. \begin{array}{l} dS_{\text{system}} = \frac{\delta q}{T} + dS_{\text{irr}} \\ \delta q = dU_{\text{system}} + \delta w \end{array} \right\} dS_{\text{system}} = \frac{dU_{\text{system}} + \delta w}{T} + dS_{\text{irr}}$$

- If the temperature remains constant throughout the process (and equal to the temperature of the reservoir supplying heat to the system), then integration from state A to state B gives

$$\delta w = TdS_{\text{system}} - dU_{\text{system}} - TdS_{\text{irr}}$$

$$\delta w \leq TdS_{\text{system}} - dU_{\text{system}}$$

$$w_{\text{max}} = q_{\text{rev}} = T(S_B - S_A) - (U_B - U_A) \quad \leftarrow w \leq T(S_B - S_A) - (U_B - U_A)$$



# Summary of Chapter 3

A heat engine is operating in a Carnot cycle: all processes are reversible. For each of the processes, obtain expressions for the change in entropy of the engine and the change in entropy of the surroundings after one cycle. Assume one mole of an ideal monatomic gas.

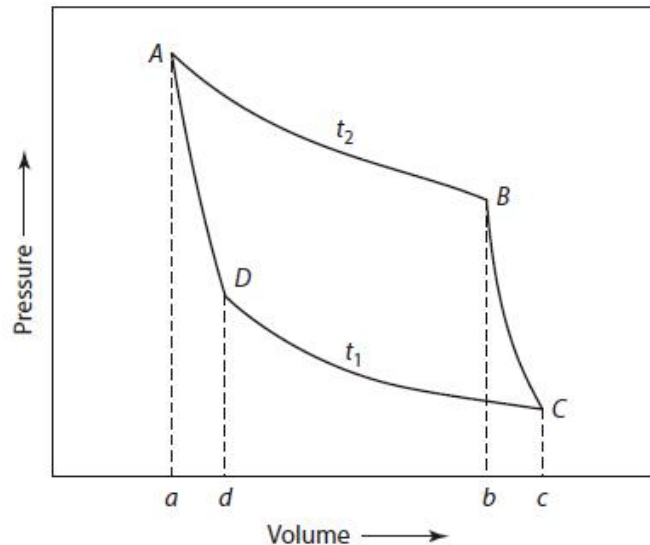
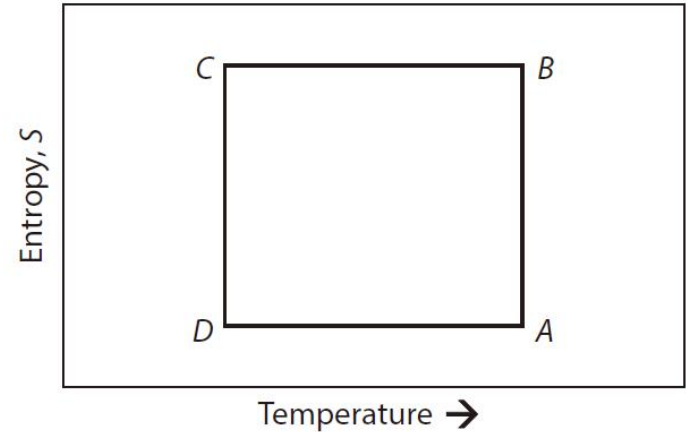


Figure 3.4 A pressure vs. volume depiction of a Carnot cycle.



# Summary of Chapter 3

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## *The engine :*

1. For the reversible isothermal expansion process from  $A$  to  $B$  :

$$\Delta S_{engine} = \frac{q_2}{T_2} = R \ln \frac{V_B}{V_A}$$

2. For the reversible adiabatic expansion process from  $B$  to  $C$  :

$$q = 0 \Rightarrow \Delta S_{engine} = 0$$

3. For the reversible isothermal compression process from  $C$  to  $D$  :

$$\Delta S_{engine} = \frac{q_1}{T_1} = R \ln \frac{V_D}{V_C}$$

4. For the reversible adiabatic compression process from  $D$  to  $A$  :

$$q = 0 \Rightarrow \Delta S_{engine} = 0$$

# Concepts and terms introduced in chapter 3

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Adiabatic expansion of gas

Carnot cycle

Constant-temperature heat reservoir

Dissipation of energy

Driving force

Efficiency of engine

Entropy

Equilibrium state

Heat engine

Irreversible processes

Isentropic process

Isothermal expansion of gas

Maximum work

Natural processes

Nonequilibrium state

Perpetual motion

Principle of Clausius

Principle of Kelvin and Planck

Quasi-static process

Reversible processes

Second Law of Thermodynamics

Spontaneous processes

## **Chapter 4 The statistical interpretation of entropy**

## 4.1 Introduction

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- ◆ Classical thermodynamics is phenomenological; that is, it deals with matter as it appears to our senses. In this regard, we use our senses to describe the properties of matter such as pressure, volume, and temperature.
- ◆ Classical thermodynamics does not delve into the more fundamental aspect, of what does matter consist? In this chapter, we will use the understanding of matter consisting of atoms and molecules and introduce the use of statistics to the approach of the topic of thermodynamics.

## 4.1 Introduction

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- ◆ In Chapter 3 the introduction of entropy as a state function was facilitated by the realization that there exist **possible and impossible processes**, and by an examination of the heat and work effects occurring during these processes.
- ◆ From the formal statement of the Second Law of Thermodynamics, as developed from Classical Thermodynamics arguments, it is difficult to assign a *physical significance or a physical quality to entropy*.
- ◆ In this respect entropy differs from internal energy in spite of the fact that, within the scope of Classical Thermodynamics, both properties are simply mathematical functions of the state of a system.

## 4.1 Introduction

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- ◆ The ready acceptance of the First Law of Thermodynamics, after its enunciation, was due to the easily understood physical significance of internal energy, whereas the lack of corresponding understanding of entropy caused the acceptance of the Second Law of Thermodynamics to be slow.
- ◆ From the classical viewpoint, the Second Law is only valid as a direct result of the fact that, to date, **human ingenuity has failed to invent a perpetual motion machine**. Thus, within the scope of Classical Thermodynamics, the Second Law is a “law” only because it has not yet been disproved. The physical interpretation of entropy had to await the development of quantum theory and statistical mechanics.

## 4.2 Entropy and disorder on an atomic scale

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- ◆ Gibbs described the entropy of a system as being a measure of its “**degree of mixed-up-ness**” at the atomic or molecular level. We can understand this concept of mixed-up-ness to be applied at the atomic or molecular level; that is, **the more “mixed up” the constituent particles of a system, the larger is the value of its entropy.**
- For example, in the **crystalline solid state** most of the constituent particles are confined to vibrate about their regularly arrayed lattice positions, whereas in **the liquid state**, confinement of the particles to lattice sites is absent and the particles are relatively free to wander through the communal volume occupied by the liquid.



## 4.2 Entropy and disorder on an atomic scale

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- The arrangement of the particles in the crystalline solid state is thus more **ordered** than that of the liquid state, or, alternatively, is less “mixed up” than that of the liquid state, and, as a consequence, the entropy of the liquid state is greater than that of the solid state.
- The atomic or molecular **disorder** in the gaseous state is greater than that in the liquid state, since, in the gaseous state, there is more volume in which the molecules are free to move. Thus, the entropy of the gaseous state is greater than that of the liquid state.

## 4.2 Entropy and disorder on an atomic scale

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- ◆ This qualitative understanding of entropy correlates with phenomena on a macroscopic level. For example, the transformation of a solid to a liquid at its equilibrium melting temperature,  $T_m$ , requires that the substance absorb a quantity of thermal energy,  $q$ , called the enthalpy of melting. The entropy of the substance being melted is thus understood to be increased. Indeed, if the melting process is conducted at constant pressure:

$$\Delta S_{\text{melting}} = \frac{\Delta H_{\text{melting}}}{T_m}$$

The increase in the entropy of the substance which accompanies melting correlates with the corresponding increase in the degree of disorder of the configurational states of its constituent particles.

## 4.2 Entropy and disorder on an atomic scale

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- ◆ Care must be taken, however, in using the concept of mixed-up-ness.
- An isolated **supercooled liquid** increases its entropy when it spontaneously transforms to a crystal, since the process is an irreversible one. But a crystal is certainly less mixed up than a liquid! How can the entropy of the substance increase?
- This apparent anomaly exists because we have focused on only one of the aspects of entropy—namely, the **configurational** aspect. If we include in our examination the influence of the enthalpy of freezing (the thermal energy released during freezing), it is seen that this release of energy increases the temperature of the isolated system, resulting in a concomitant increase in the **thermal entropy** of the system.
- This increase in thermal entropy is greater than that of the decrease in configurational entropy during the liquid to crystal transformation. This shows that all aspects of the entropy of a system must be considered when evaluating changes in a system.

## 4.2 Entropy and disorder on an atomic scale

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- ◆ If the transformation from liquid to solid occurs isothermally at the equilibrium melting temperature of the substance,  $T_m$ , then the increase in the degree of disorder of the heat reservoir equals the decrease in the degree of disorder of the substance, and the total degree of disorder of the combined system plus heat reservoir is unchanged. Consequently, the entropy of the combined system is unchanged as a result of the ***equilibrium freezing process***; entropy has been transferred from the substance to the heat reservoir.
- ◆ **The equilibrium melting or freezing temperature of a substance** can thus be defined as that temperature at which no change in the entropy of the substance plus heat reservoir occurs as a result of the phase change. Only at this temperature are the solid and liquid in equilibrium with one another, and hence, only at this temperature can the phase change occur reversibly, with no net increase of entropy.

## 4.3 The concept of microstate

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- ◆ In classical thermodynamics, the *state* of a single-component isolated system, which in this chapter is denoted as its *macrostate*, is fixed with the knowledge of two of its thermodynamic variables, usually considered to be the intensive thermodynamic variables of pressure and temperature.
- ◆ When we consider the atomic and molecular makeup of the system, there are many more possible configurations of the constituents which give rise to the same macrostate of the system. Each constituent has three coordinates for its position and three coordinates for its momentum. In addition, there is an enormous number of constituents in real systems, of the order of  $10^{24}$  for a mole of particles.

## 4.3 The concept of microstate

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- ◆ Statistical thermodynamics considers the various ways that the total number of constituents,  $N$ , can be configured in the systems we encountered in the first three chapters, namely:
  1. Isolated systems, where  $S=S(U,V,N)$  or  $U=U(S,V,N)$
  2. Closed systems in equilibrium with a heat bath, where  $S=S(T,V,N)$
  3. Open systems, where  $S=S(T,V,\mu)$
- ◆ When considering such systems, special names have been given in the statistical approach to thermodynamics— namely, the microcanonical, canonical, and grandcanonical formalisms, respectively. In this brief introduction to statistical thermodynamics, we will consider the microcanonical (isolated) cases.

## 4.3 The concept of microstate

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- ◆ The development of a quantitative relationship between entropy and the degree of mixed-up-ness requires quantification of the term degree of mixed-up-ness, and this can be obtained from a consideration of elementary statistical thermodynamics. Statistical thermodynamics postulates that the equilibrium state of a system is simply the most probable of all of its possible (i.e., accessible) microstates. Therefore, statistical thermodynamics is concerned with:
  1. The determination of the most probable microstate
  2. The criteria governing the most probable microstate
  3. The properties of this most probable microstate
- ◆ Both Boltzmann and Gibbs found it convenient to examine the distribution of energies among the particles of the system by placing the energy of the particles into discrete compartments.

## 4.3 The concept of microstate

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- ◆ This changed the distribution of energy of the particles from being a **continuous function of state** to a **discrete function** and therefore made it more convenient to perform various statistical operations on the system of interest. This method turned out to be similar (but not identical) to the one used by quantum theory, which was developed several decades later. A postulate of quantum theory is that, if a particle is confined to move within a given fixed volume, then its energy is quantized; that is, the particle may only have certain discrete values of energy, which are separated by **forbidden energy bands**.
- ◆ For any given particle, the spacing between the quantized values of energy (the allowed energy levels) decreases as the volume available to the movement of the particle increases. This identifies another aspect of the entropy of a system—namely, **the degree of spread** in the distribution of the particles among **its possible energy levels**. This spread can be considered the degree of mixed-up-ness of the occupied energy levels of the particles in the system and is related to the system's thermal entropy.



## 4.4 The microcanonical approach

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### 4.4.1 Identical Particles on *Distinguishable* Sites with Different Assigned Energies

- The effect of the quantization of energy and its resulting distribution can be illustrated by considering a hypothetical system comprising a **perfect crystal** in which all of the distinguishable sites are occupied by *identical* particles.
- The characteristics of the particles and the crystal structure determine the quantization of the allowed energy levels, in which the lowest energy level, or the **ground state**, is designated  $\epsilon_0$ , and the succeeding levels of increasing energy are designated  $\epsilon_1$ ,  $\epsilon_2$ ,  $\epsilon_3$ , and so on.

## 4.4 The microcanonical approach

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### 4.4.1 Identical Particles on *Distinguishable* Sites with Different Assigned Energies

- ◆ The crystal contains  $n$  particles and has the fixed energy  $U$  and fixed volume  $V$ . The system is therefore considered to be an *isolated* one: neither energy nor particles can enter or leave the system. Statistical thermodynamics asks the following questions:
  - In how many ways can the  $n$  particles be distributed over the available energy levels such that the total energy of the crystal (i.e.,  $U$ ) remains the same?
  - Of the possible distributions, which is the most probable?

## 4.4 The microcanonical approach

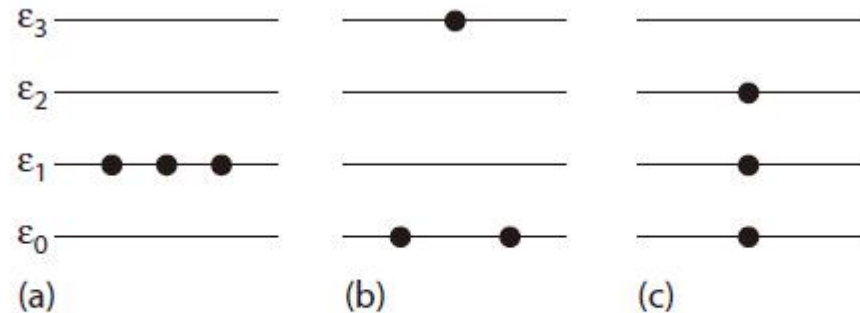
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### 4.4.1 Identical Particles on *Distinguishable* Sites with Different Assigned Energies

- ◆ Consider that the crystal contains three identical particles which are located on three distinguishable lattice sites  $A$ ,  $B$ , and  $C$ . Suppose, for simplicity, that the quantization is such that the energy levels are equally spaced, with the ground level being taken as  $\varepsilon_0=0$ , the first level  $\varepsilon_1=u$ , the second level  $\varepsilon_2=2u$  and so on, and let the total energy of the system be  $U=3u$ .

## 4.4 The microcanonical approach

### 4.4.1 Identical Particles on *Distinguishable* Sites with Different Assigned Energies



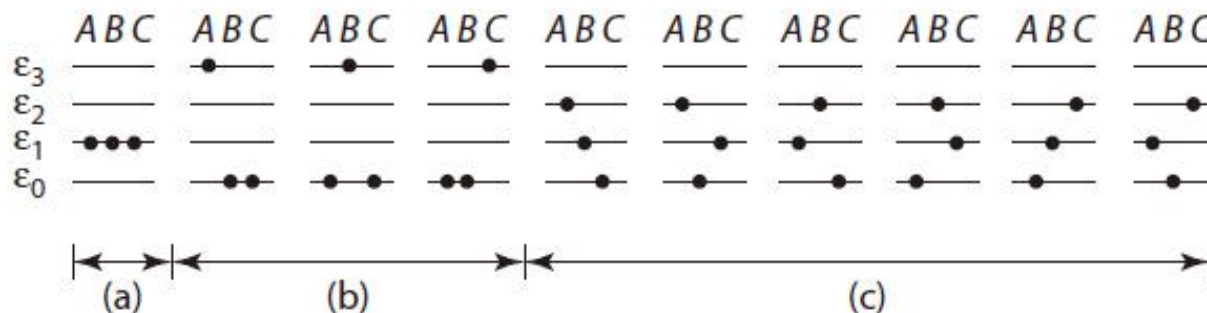
**Figure 4.1** The distributions of particles among energy levels in a system of constant energy.

- a. All three particles in level 1
- b. One particle in level 3, and the other two particles in level 0
- c. One particle in level 2, one particle in level 1, and one particle in level 0

## 4.4 The microcanonical approach

### 4.4.1 Identical Particles on *Distinguishable* Sites with Different Assigned Energies

- *Distribution a* . There is **only one** microstate of this distribution, since the interchange of the particles among the three sites does not produce a different microstate.

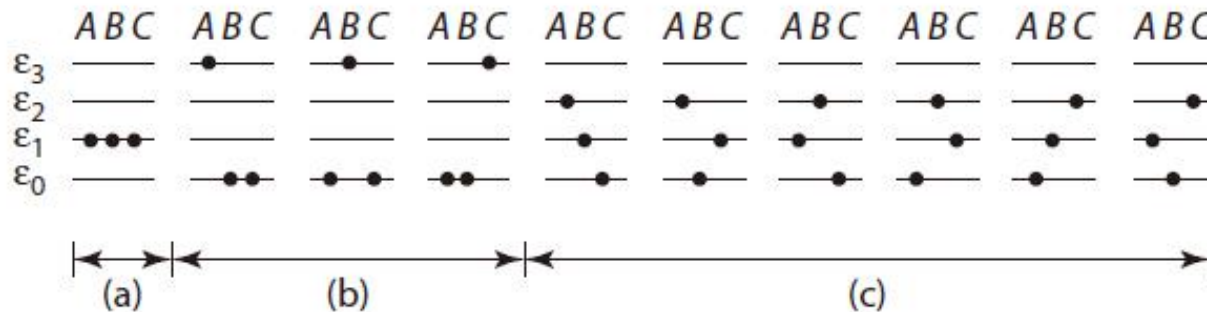


**Figure 4.2** Illustration of the complexions or microstates within distributions of particles among energy levels in a system of constant energy.

## 4.4 The microcanonical approach

### 4.4.1 Identical Particles on *Distinguishable* Sites with Different Assigned Energies

- *Distribution b.* Any of the three distinguishable sites can be occupied by any of the three particles of energy  $3u$ , and the remaining two sites are each occupied by a particle of zero energy. Since the interchange of the particles of zero energy does not produce a different arrangement, there are **three** microstates in distribution *b*.

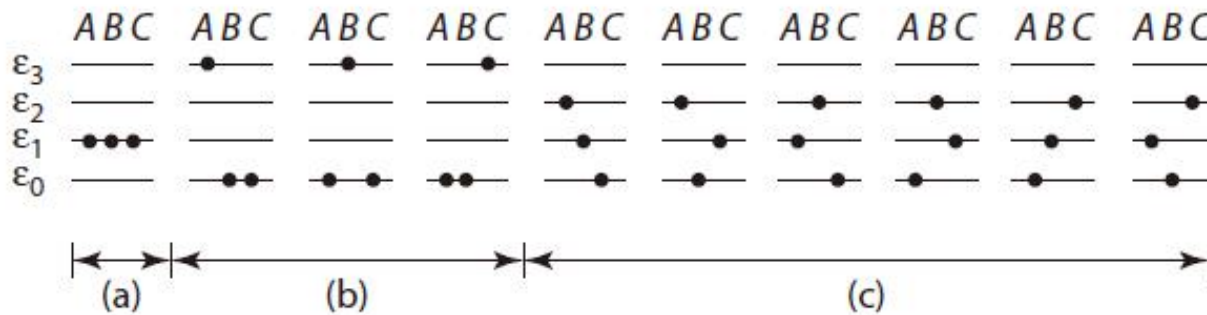


**Figure 4.2** Illustration of the complexions or microstates within distributions of particles among energy levels in a system of constant energy.

## 4.4 The microcanonical approach

### 4.4.1 Identical Particles on *Distinguishable* Sites with Different Assigned Energies

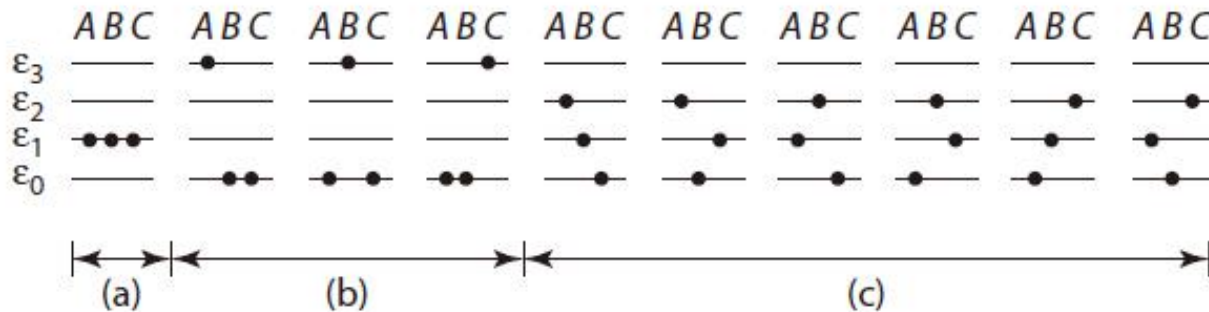
• *Distribution c* . Any of the three distinguishable sites can be occupied by the particle of energy  $2u$ . Either of the two remaining sites can be occupied by the particle of energy  $1u$ , and the single remaining site is occupied by the particle of zero energy. The number of distinguishable microstates in distribution *c* is thus  $3 \times 2 \times 1 = 3! = 6$ .



**Figure 4.2** Illustration of the complexions or microstates within distributions of particles among energy levels in a system of constant energy.

## 4.4 The microcanonical approach

### 4.4.1 Identical Particles on *Distinguishable* Sites with Different Assigned Energies



**Figure 4.2** Illustration of the complexions or microstates within distributions of particles among energy levels in a system of constant energy.

The 10 arrangements are the 10 microstates of the system. Since each of the *microstates* have the same energy, they correspond to one *macrostate* of the system.



## 4.4 The microcanonical approach

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### 4.4.1 Identical Particles on *Distinguishable* Sites with Different Assigned Energies

- ◆ The concept of macrostate lies within the domain of classical thermodynamics. The macrostate is fixed when the values of the independent variables are fixed. In the preceding example, the system was considered to be isolated; that is, the values of  $U$ ,  $V$ , and  $N$  are fixed (the constancy of volume is required in order that the quantization of the energy levels can be determined), and hence, the macrostate of the system is fixed.
- ◆ With respect to the occupancy of the microstates within any macrostate, in view of the absence of any reason to the contrary, it is assumed that each of the microstates is equally probable, and thus, the probability of observing the preceding system in any one of its 10 possible microstates is  $1/10$ .

## 4.4 The microcanonical approach

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### 4.4.1 Identical Particles on *Distinguishable* Sites with Different Assigned Energies

◆ The ten microstates occur in three distinct distributions, and hence, the probability that the system occurs.

- In distribution *a* is  $1/10$
- In distribution *b* is  $3/10$
- In distribution *c* is  $6/10$

Distribution *c* is thus the most probable .

## 4.4 The microcanonical approach

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### 4.4.1 Identical Particles on *Distinguishable* Sites with Different Assigned Energies

Distribution  $c$  is thus the most probable .

1. If it were possible to make an instantaneous observation of the system, the probability of observing an arrangement in distribution  $c$  would be  $6/10$ .
2. If the system were observed over a finite interval of time, during which the system rapidly changed from one microstate to another, the fraction of time which the system spends in all of the arrangements in distribution  $c$  would be  $6/10$ .

## 4.4 The microcanonical approach

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### 4.4.1 Identical Particles on *Distinguishable* Sites with Different Assigned Energies

- ◆ As the total energy of the system and the number of particles which it contains increase, the number of distinguishable microstates also increases, and, for given values of  $U$ ,  $V$ , and  $N$ , these microstates still correspond to a single macrostate.
- ◆ Similarly, the number of possible distributions increases, and in real systems—for example, 1 mole of a system which contains  $6.023 \times 10^{23}$  particles—the number of arrangements within the most probable distribution is very much larger than the number of arrangements in all of the other distributions ( $\Omega \approx \Omega_{\max}$ ).
- ◆ The most probable microstate is considered to be *the* equilibrium state in the macro-system, and the total number of possible arrangements,  $\Omega$ , is set equal to the number of microstates with maximum probability,  $\Omega_{\max}$ .

## 4.4 The microcanonical approach

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### 4.4.1 Identical Particles on *Distinguishable* Sites with Different Assigned Energies

- ◆ Since the equilibrium state in a closed system is the one of greatest entropy, the entropy,  $S$ , varies with  $\Omega$ :

$$S \sim \Omega$$

Hence, the entropy of an isolated system is calculated from the microstates which maximize the spread of the particles among the energy levels. From the preceding simplified example, the maximum entropy occurs when the occupied energy levels of the particles are the most spread out.

## 4.4 The microcanonical approach

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### 4.4.2 Configurational Entropy of Differing Atoms in a Crystal

- ◆ In the preceding discussion, the entropy was considered in terms of the number of ways that the energy of identical but distinguishable particles could be distributed.
- ◆ The entropy was seen to be greatest when the occupancy of the energy levels was broadly distributed. The degree of mixed-up-ness corresponded to the spread of the distribution.
- ◆ Entropy can also be considered in terms of the number of ways in which particles themselves can be distributed in space, and this consideration gives rise to the concept of *configurational entropy*.

## 4.4 The microcanonical approach

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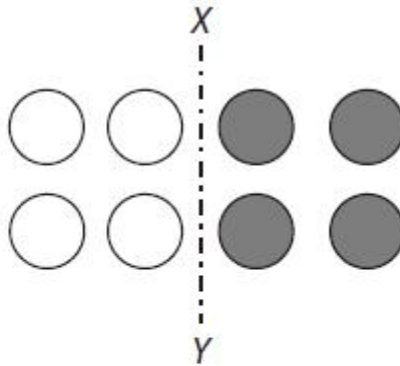
### 4.4.2 Configurational Entropy of Differing Atoms in a Crystal

- ◆ Consider two crystals, one containing white atoms and the other containing gray atoms. Assume that there is no difference in the energy of white/white, white/gray, and gray/gray bonds. When the two crystals are placed in physical contact with one another, a *spontaneous process* occurs in which the white atoms diffuse into the crystal of the gray atoms and the gray atoms diffuse into the crystal of the white atoms.
- ◆ Since this process is spontaneous, entropy must be produced. If the system is isolated, it is predicted that equilibrium will be reached when the entropy of the system reaches a maximum value, which would be *when the diffusion processes have occurred to the extent that all concentration gradients in the system have been eliminated*. This is the mass transport analog of the heat transfer case, in which heat flows irreversibly between two bodies until the temperature gradients have been eliminated.

## 4.4 The microcanonical approach

### 4.4.2 Configurational Entropy of Differing Atoms in a Crystal

- ◆ Consider the two crystals to be composed of four white atoms and four gray atoms, which shows the initial arrangement of the atoms. Assume the crystals are held at the same temperature and volume. As stated previously, we assume that there are no energetically favored bonds.



**Figure 4.3** Representation of a crystal of white atoms in contact with a crystal of gray atoms.



## 4.4 The microcanonical approach

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### 4.4.2 Configurational Entropy of Differing Atoms in a Crystal

- ◆ If the partition is removed, the atoms are able to rearrange themselves by diffusion. We seek to determine the most probable state of the system; that is, where will the atoms reside with respect to the position of the barrier after equilibrium is attained (i.e., how many gray atoms will be on the left side and how many will be on the right side of the system)?
- ◆ The use of combinatorics allows us to calculate the number of distinguishable ways in which this arrangement can be realized. Recall that the number of ways that  $N$  atoms can be arranged into two groups of  $n$  atoms each is given as

$$\Omega = \frac{N!}{n! \times (N - n)!}$$

## 4.4 The microcanonical approach

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### 4.4.2 Configurational Entropy of Differing Atoms in a Crystal

◆ First, consider the number of possible distributions of white atoms on the left side:

1. Four white atoms on the left:  $\Omega_{4,0} = (4 \times 3 \times 2 / 4! 0!) = 1$  (in which the notation indicates four white atoms on the left of  $XY$  and none on the right).
2. Three white atoms on the left:  $\Omega_{3,1} = (4! / 3! 1!) = 4$  (the “missing” white atom could be in any of the four sites).
3. Two white atoms on the left:  $\Omega_{2,2} = (4! / 2! 2!) = 6$ .
4. One white atom on the left:  $\Omega_{1,3} = (4! / 1! 3!) = 4$  (the one white atom could be on any of the four sites).
5. No white atoms on the left:  $\Omega_{0,4} = (4! / 0! 4!) = 1$ .

## 4.4 The microcanonical approach

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### 4.4.2 Configurational Entropy of Differing Atoms in a Crystal

- ✓ Consider the number of ways that each of the preceding configurations can be made.
- 1. All white atoms on the left and all gray atoms on the right:  $\Omega_{4,0} \times \Omega_{0,4} = 1$
- 2. Three white atoms on the left and one gray atom on the right:  
 $\Omega_{3,1} \times \Omega_{1,3} = 16$
- 3. Two white atoms on the left and two gray atoms on the right:  
 $\Omega_{2,2} \times \Omega_{2,2} = 36$
- 4. One white atom on the left and three gray atoms on the right is the same as three white atoms on the left and one gray atom on the right:  
 $\Omega_{1,3} \times \Omega_{3,1} = \Omega_{3,1} \times \Omega_{1,3} = 16$
- 5. No white atoms on the left can only be done one way:  $\Omega_{0,4} \times \Omega_{4,0} = 1$

## 4.4 The microcanonical approach

### 4.4.2 Configurational Entropy of Differing Atoms in a Crystal

| Atoms on Left of Partition | Atoms on Right of Partition | No. of Ways (Complexions) |
|----------------------------|-----------------------------|---------------------------|
| 4 white                    | 4 gray                      | 1                         |
| 3 white/1 gray             | 3 gray/1 white              | 16                        |
| 2 white/2 gray             | 2 gray/2 white              | 36                        |
| 1 white/3 gray             | 1 gray/3 white              | 16                        |
| 4 gray                     | 4 white                     | 1                         |

## 4.4 The microcanonical approach

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### 4.4.2 Configurational Entropy of Differing Atoms in a Crystal

- ◆ The total number of spatial configurations available to the system is  $1+16+36+16+1=70$ , which is the number of distinguishable ways in which four atoms of one color and four atoms of the other color can be arranged on eight sites; that is,

$$\text{number of ways} = \frac{8!}{4!4!} = 70$$

If, as before, it is assumed that each of these configurations is equally probable, then the probability of finding the system in the arrangements 4:0 or 0:4 is  $1/70$ , the probability of arrangements 3:1 or 1:3 is  $16/70$ , and the probability of arrangement 2:2 is  $36/70$ . Arrangement 2:2 is thus the most probable and thus corresponds to the equilibrium state, in which the concentration gradients have been eliminated.

## 4.4 The microcanonical approach

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### 4.4.2 Configurational Entropy of Differing Atoms in a Crystal

- ◆ The most likely arrangement of atoms would be to have two gray atoms and two white atoms on both sides of the permeable partition. But other arrangements are possible, and since there is no energetic difference between the various microstates, all are equally probable.
- ◆ However, since there are 36 microstates with equal numbers of gray and white atoms on either side, that macrostate would be most probable. Once again, the most probable case is the one with the greatest configurational entropy for the given energy.
- ◆ Statistical thermodynamics does not determine with absolute certainty what the final arrangement will be, but only that which is the most probable configuration and the most likely to be observed. Again, it is emphasized that for large systems, there is negligible chance for the system to be found away from its maximum  $\Omega$ .

## 4.4 The microcanonical approach

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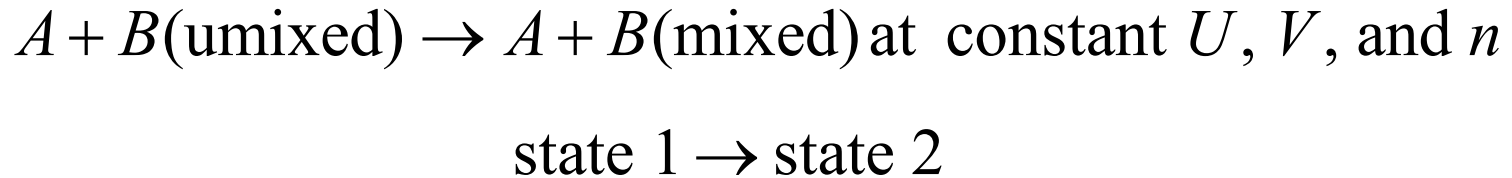
### 4.4.2 Configurational Entropy of Differing Atoms in a Crystal

- ◆ In the preceding case, the increase in entropy occurs as a result of the increase in the number of spatial configurations which become available to the system when the crystals of  $A$  and  $B$  are placed in contact with one another. The increase in the entropy of the system arises from an increase in its configurational entropy,  $S_{\text{conf}}$ .
- ◆ It must be kept in mind that this is for the case where there is **no energetically favored bonds** between or among the particles of the system.
- ◆ If, for example, the white/gray bond energy was much greater than the white/white and gray/gray bond energy, the system may have to remain in order to keep the total energy of the system constant.

## 4.4 The microcanonical approach

### 4.4.2 Configurational Entropy of Differing Atoms in a Crystal

The mixing process can be expressed as



$$\begin{aligned}\Delta S'_{\text{conf}} &= S_{\text{conf}(2)} - S_{\text{conf}(1)} = k_B \ln(\Omega_{\text{conf}(2)}) - k_B \ln(\Omega_{\text{conf}(1)}) \\ &= k_B \ln \frac{\Omega_{\text{conf}(2)}}{\Omega_{\text{conf}(1)}}\end{aligned}$$

If  $n_a$  atoms of  $A$  are mixed with  $n_b$  atoms of  $B$  :

$$\Omega_{\text{conf}(2)} = \frac{(n_a + n_b)!}{n_a! n_b!}$$

$$\Delta S'_{\text{conf}} = k_B \ln \frac{(n_a + n_b)!}{n_a! n_b!}$$
$$\Omega_{\text{conf}(1)} = 1$$



## 4.4 The microcanonical approach

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### 4.4.2 Configurational Entropy of Differing Atoms in a Crystal

If we consider  $X_A$  and  $X_B$  to be the mole fraction of  $A$  atoms and the mole fraction of  $B$  atoms, the expression can be reduced to

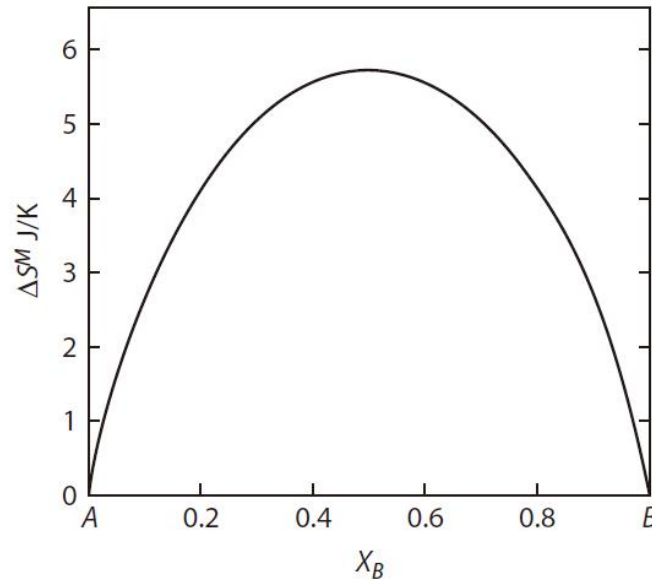
$$\Delta S_{conf} = -R (X_A \ln X_A + X_B \ln X_B)$$

for 1 mole of atoms. This equation gives the configurational entropy of a binary system containing two distinguishable kinds of atoms.

$$\ln N! \approx N \ln N - N$$

## 4.4 The microcanonical approach

### 4.4.2 Configurational Entropy of Differing Atoms in a Crystal



**Figure 4.4** The molar configurational entropy of mixing for a binary solution with no preference for any type of bonds. The maximum value is  $R \ln 2$ .

The maximum configurational entropy of mixing can be seen to occur when equal numbers of the two kinds of atoms are mixed together to form a (solid) solution.

## 4.4 The microcanonical approach

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### 4.4.3 Configurational Entropy of Magnetic Spins on an Array of Atoms

- ◆ Consider a collection of magnetic spins in a paramagnet, i.e., one in which the spins are randomly placed on the atoms. The spins may take values of  $\pm 1/2$ . The spins will be called *up* and *down* spins for convenience, even though they do not have a direction in space.
- ◆ If there is no external magnetic field applied and if the interaction energy between the spins (also called the *exchange energy*) is nonexistent, each of the two quantum states (up or down) has the same energy.
- ◆ This is therefore another case of the microcanonical formalism, since the total energy of the system is the sum of the individual energies of the spins and remains constant. We wish to enumerate all possible microstates of the system.

## 4.4 The microcanonical approach

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### 4.4.3 Configurational Entropy of Magnetic Spins on an Array of Atoms

- ◆ Consider a system consisting of eight sites on which magnetic spins will be placed. One microstate is that all atoms have their spins pointing up. There is only **one** such microstate state possible.
- ◆ Consider a microstate with one down spin. The down spin could be on any of the eight sites. Hence, there are **eight** such possible microstates with one down spin.
- ◆ Consider a microstate with two down spins. The first one could be on any of the eight atoms and the second could be on any of the remaining seven atoms. This would be **56** microstates, except that we have counted the states twice, since the order in which we placed the spins on the atoms could be switched without producing a new microstate. Hence, there are 28 microstates with two down spins. This can be calculated from the following formula:

$$\text{6 up and 2 down} = 8! / 6! 2! = 28$$

*Thermodynamics of Materials & Phase transformations*

## 4.4 The microcanonical approach

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### 4.4.3 Configurational Entropy of Magnetic Spins on an Array of Atoms

The case for three down spins can be calculated from

$$5 \text{ up and } 3 \text{ down} = \frac{8!}{5! 3!} = 56$$

The case for four up and four down can be determined as

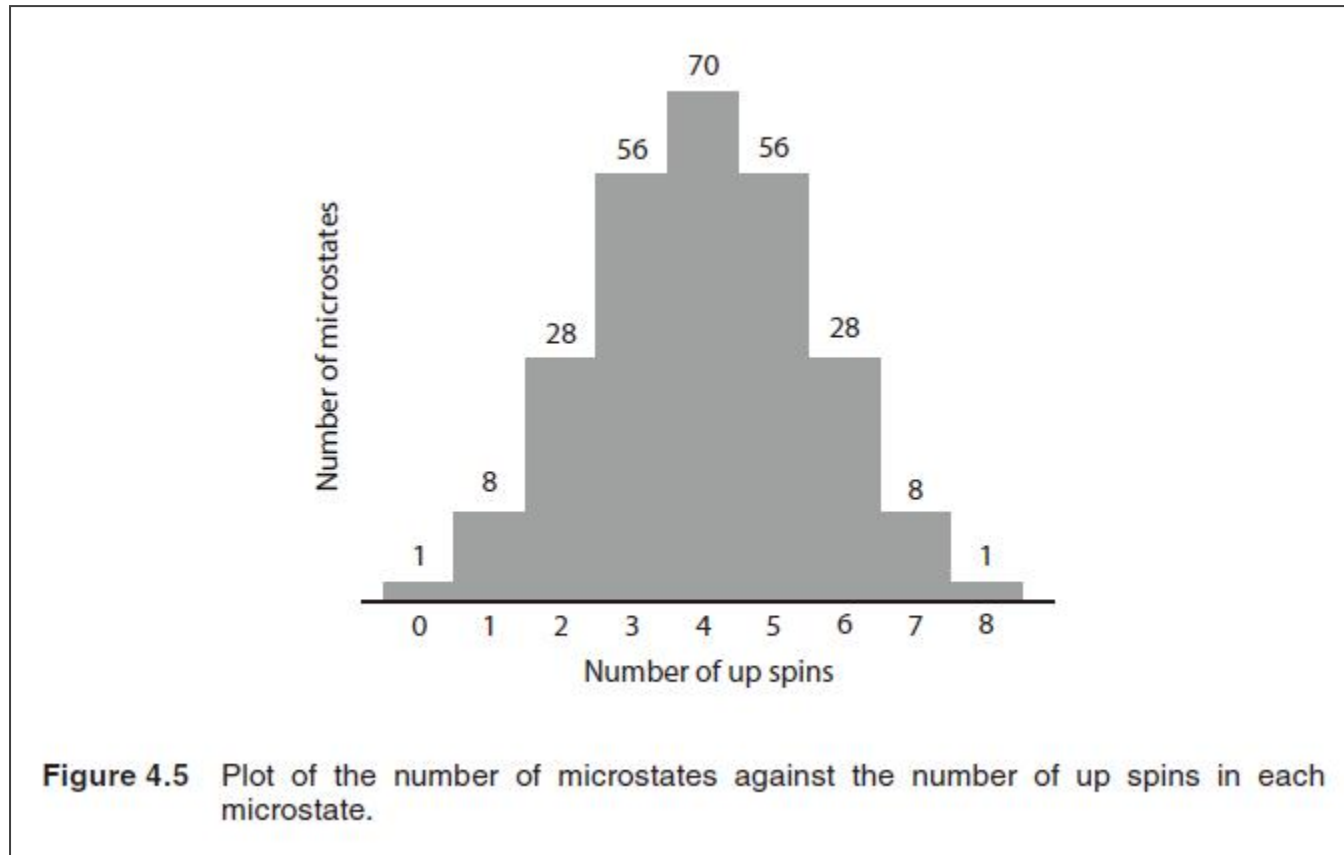
$$4 \text{ up and } 4 \text{ down} = \frac{8!}{4! 4!} = 70$$

The other cases can easily be seen by symmetry to be

- 3 up and 5 down = 56
- 2 up and 6 down = 28
- 1 up and 7 down = 8
- 0 up and 8 down = 1

## 4.4 The microcanonical approach

### 4.4.3 Configurational Entropy of Magnetic Spins on an Array of Atoms

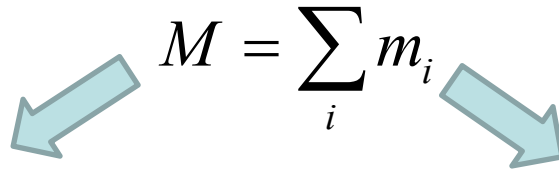


There are a total of  $2^8 = 256$  possible microstates of the spins, which can be divided into nine distinct groups

## 4.4 The microcanonical approach

### 4.4.3 Configurational Entropy of Magnetic Spins on an Array of Atoms

The magnetization of a material is defined as

$$M = \sum_i m_i$$


The magnetization

The sum of the moments of the  $\pm$  spins

It can be seen that  $M$  can be less than zero, equal to zero, or greater than zero, depending on the microstate. However, the most probable microstate has a total magnetization of zero, and this is considered to be the equilibrium state. It can also be seen that the average value of  $M$  over all microstates is also zero. A material in this state is called a *paramagnet*. If an external magnetic field is applied to the system,  $M$  would change to a nonzero value.

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*The End*

